

Ammar Asad

Islamabad, Pakistan · +92 337 7131367 · m.ammarrwarraich@gmail.com
github.com/mammarrwarraich · linkedin.com/in/ammarr-warraich-1354ba373

Education

National University of Sciences and Technology (NUST)
Bachelors in Electrical Engineering

Islamabad, Pakistan
2025 – Present

• **Current CGPA:** 3.82 / 4.00

Position: Elected Class Representative (CR) for the EE Cohort

FSc Pre-Engineering
Matriculation

Score: 93%
Score: 96%

International & National Honors

- **Highest Ranked Silver Medalist – INSO (2025):** Ranked 9th globally at the International Nuclear Science Olympiad; secured the sole international silver medal for Pakistan after a rigorous multi-stage national selection process.
- **Pride of Pakistan Nominee (2025):** Nominated by the Government of Pakistan for Independence Day Honors for outstanding achievements in scientific disciplines and global representation.
- **Pakistan's Mathematics Olympiad (2024):** Achieved a national rank of 23rd out of nationwide competitors in the challenge jointly organized by FBISE and LUMS.
- **Copernicus Physics Olympiad (2023):** Awarded an international Honorable Mention for theoretical physics problem-solving.
- **Open Cybersecurity Challenge:** Attained National Semi-Finalist standing in the university-level security infrastructure tracking competition.

Technical Toolbelt

- **Proficient:** C++, Python, Arduino IDE, Embedded C
- **Design & Simulation Software:** MATLAB, Simulink, LTspice, AutoCAD
- **Lab Bench Tools:** Oscilloscopes, Function Generators, Digital Multimeters (DMM)
- **Version Control & Utilities:** Git, GitHub, L^AT_EX, Markdown

Engineering Projects

BodePlotter Pro · C++, Hardware Automation

Continuous Development

- Built a C++ automated testing tool interfacing with hardware instruments to execute automated frequency sweeps ($10Hz \sim 1MHz$), cutting manual circuit characterization time by $\sim 80\%$.
- Selected the Durand-Kerner algorithm over Newton's method for simultaneous polynomial root-finding to extract transfer function poles reliably from measured impedance data.
- Utilized Horner's Method to optimize real-time floating-point arithmetic execution loops, minimizing processing latency on the microcontroller.

Automatic Power Factor Correction (APFC) Meter · Embedded C++, Smart Grid Systems

May 2026

- Designed and fabricated an Arduino-based smart monitoring unit using automated relay modules to dynamically switch localized capacitor steps.
- Authored real-time firmware loops to track current-voltage phase deltas and compute instantaneous reactive power shifts across inductive loads.
- Successfully corrected lagging power factors from ~ 0.72 up to an efficient target benchmark of 0.95 during hardware load testing.

PyDIPS (Machine Learning Network Security Suite) · Python, scikit-learn, Cybersecurity

Late 2025

- Developed a localized network security pipeline using custom Python socket-sniffing scripts to capture and log raw packet features.

- Trained a scikit-learn Random Forest classifier on labeled traffic datasets, serializing the output as a `.pkl` model for real-time inference and anomaly detection.
- Wrote a multi-threaded network attack simulator to stress-test the pipeline and validate system response times under simulated high-traffic threat vectors.

Industrial Solar Farm Layout & Design · *AutoCAD, Infrastructure Planning*

Academic Project

- Drafted AutoCAD structural footprints, structural configurations, and spatial array layouts for a conceptual $\sim 5MW$ photovoltaic installation following industrial layout constraints.
- Calculated row-pitch dimensions and angular spacing to completely eliminate winter solstice shading losses, maximizing annual solar tracking yield.
- Mapped electrical single-line routing diagrams covering string configurations, central inverter rooms, a $1000kVA$ step-up transformer, and battery storage (BESS) racks.

Leadership & Experience

Elected Class Representative (CR) · *NUST SEECS*

2026 – Present

- Serve as the primary operational link between university department heads, faculty, and an undergraduate cohort; coordinate logistics, student schedules, and departmental administrative requests.

IEEE NUST Student Branch Member · *Publications Department*

2026 – Present

- Draft, review, and curate technical content, project summaries, and research highlights for the engineering student body.

STEM Youth Lead · *PIEAS Collaboration*

2025

- Organized scientific community outreach frameworks and served as a public liaison to promote youth engagement across technical STEM disciplines.

Autonomous Drone Systems Training · *Rawalpindi Technology Park*

2024

- Completed practical hardware training in autonomous drone configurations; led a team at the 18th National Design Build and Fly Competition (NDBFC) at GIK Institute.